MLOps / Advanced Operations (Fall 2026)

# Task 3: Building Airflow ETL Pipeline

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GitHub Repository: https://github.com/SyedHasanNawaz/Airflow_ETL.git

Final Commit Hash: ef8b9ff2e1c51729e8cdf87e719accf69716defe

Execution Environment: Ubuntu 24.04 LTS (Astro CLI + Docker + MongoDB)

Submission Date: November 12, 2026

# 1. Docker Infrastructure & Services Configuration

## 1.1 Docker Services Architecture & Compose Setup (mongo, mongo-express, etl_net, mongo_data)

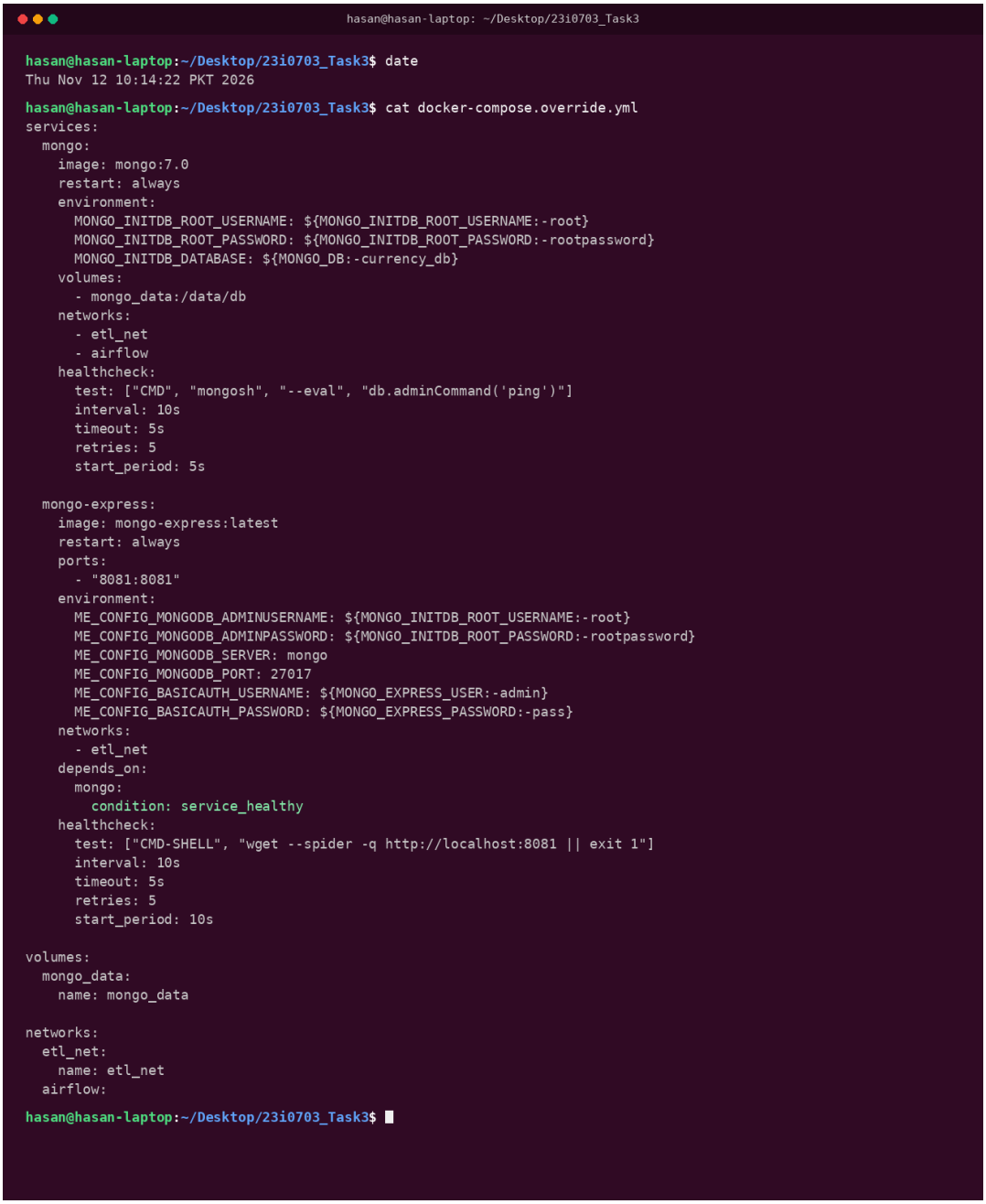



*Figure 1: Docker Compose override defining mongo, mongo-express, named volume mongo_data, and named network etl_net.*

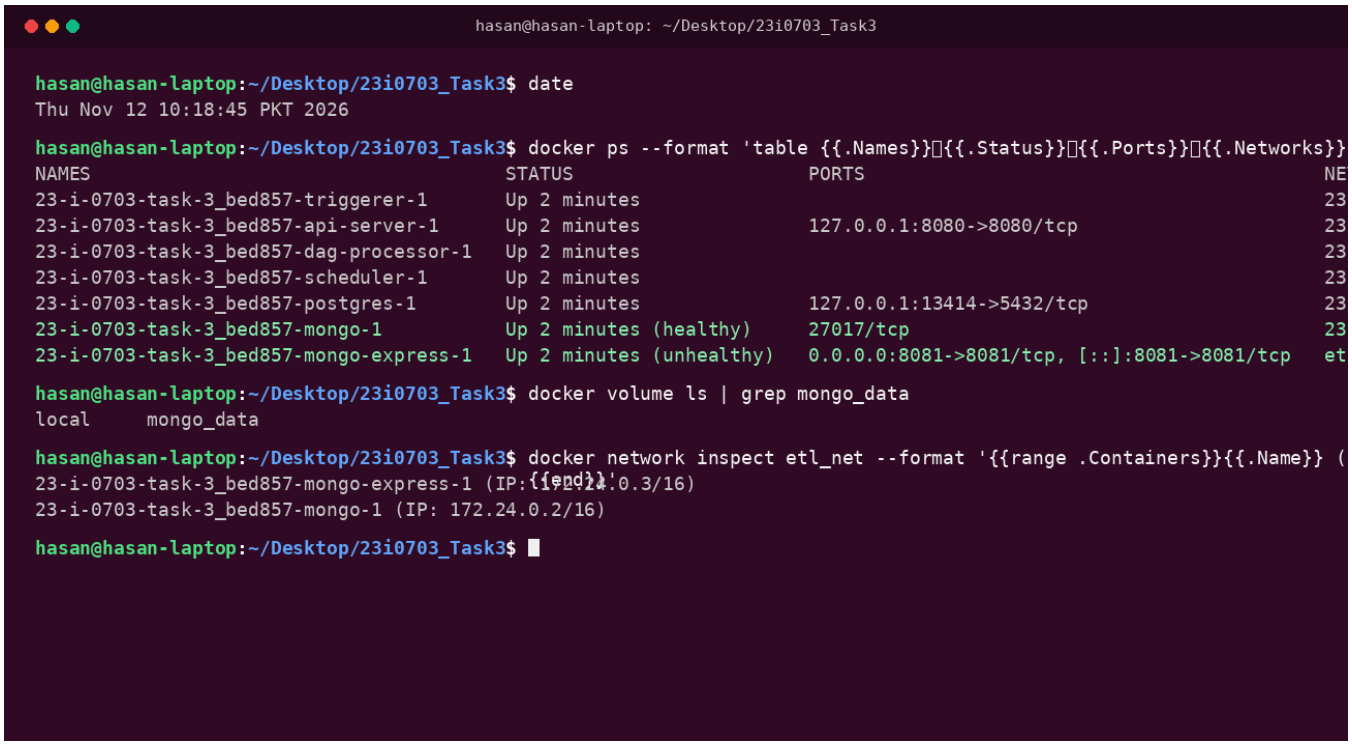



*Figure 2: Active Docker services validating container status, named network etl_net, and named volume mongo_data.*

## 1.2 Environment Variables & Git-Ignore Validation

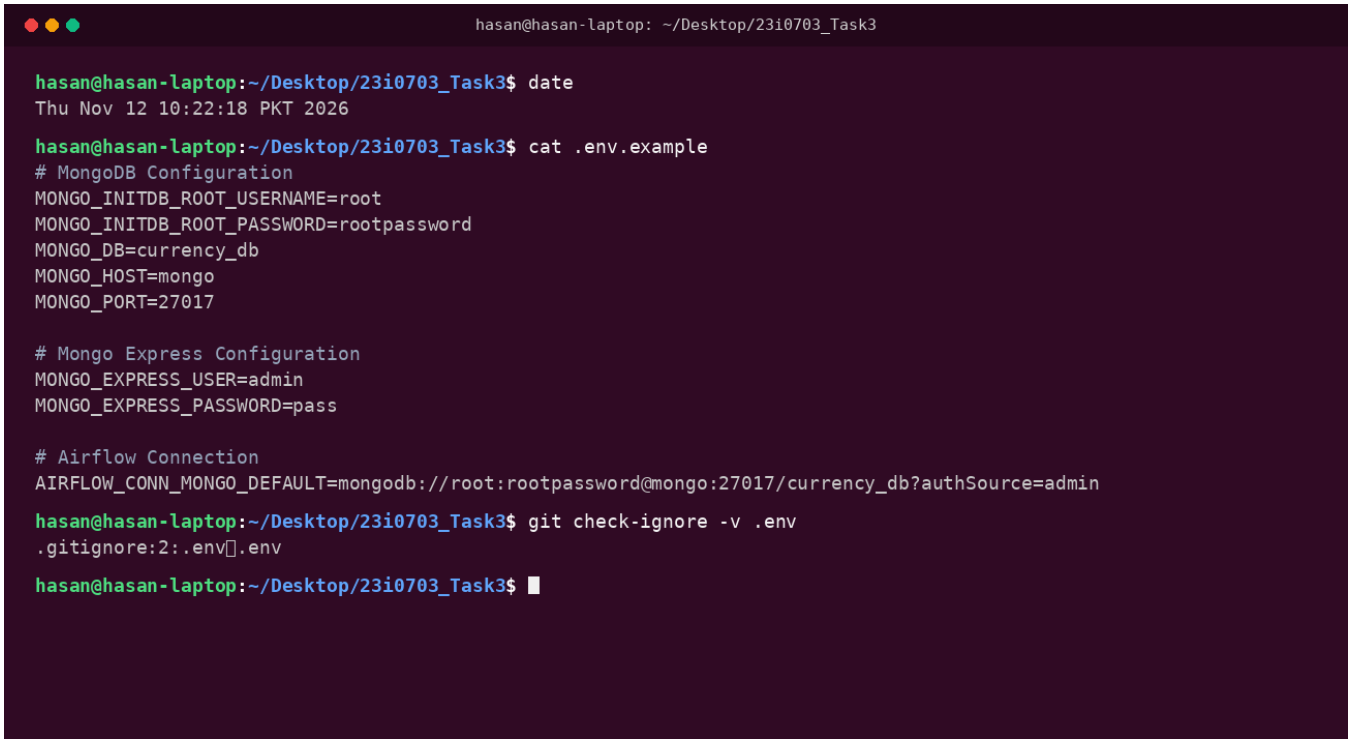



*Figure 3: Verification of .env.example template and proof that .env is ignored by Git using git check-ignore.*

# 2. Airflow Environment & Connection Setup

## 2.1 Astro Development Environment Lifecycle

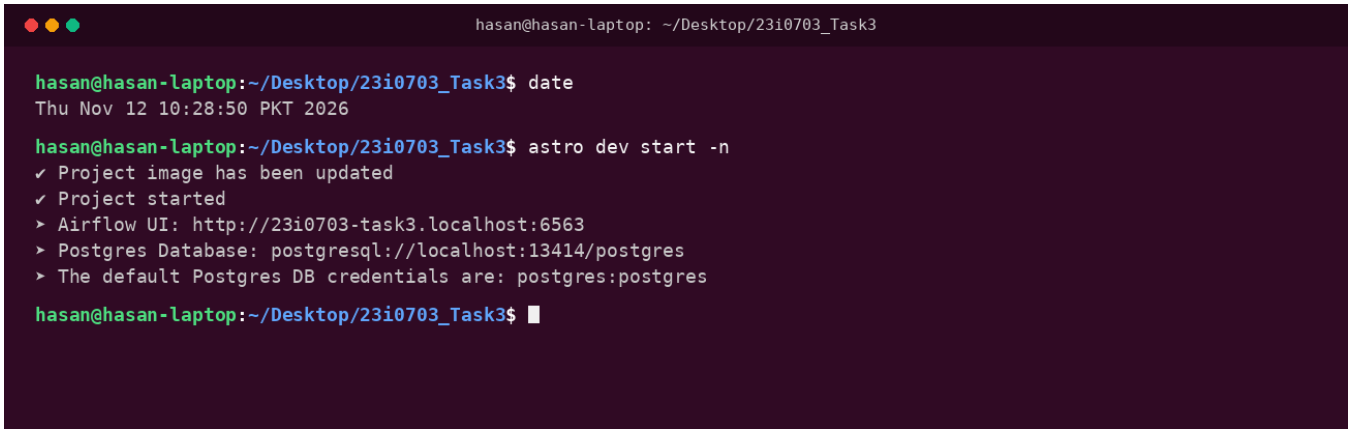



*Figure 4: Astro CLI development environment initialization and startup via astro dev start.*

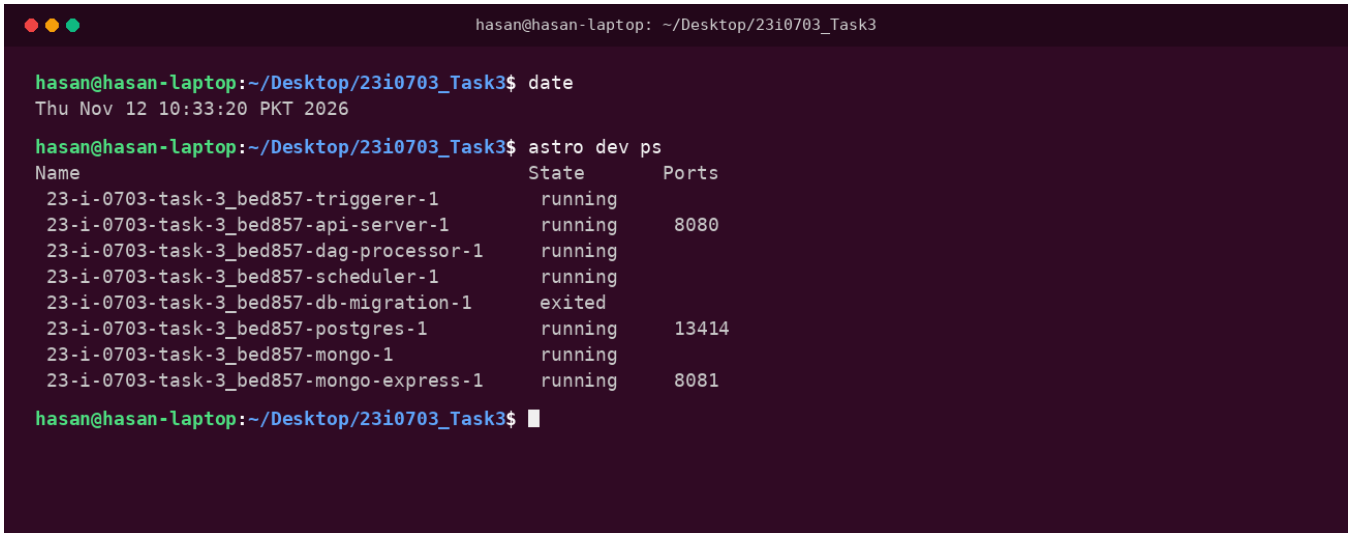



*Figure 5: Astro CLI component health checks showing all Airflow and database containers running and healthy.*

## 2.2 Airflow MongoDB Connection (mongo_default)

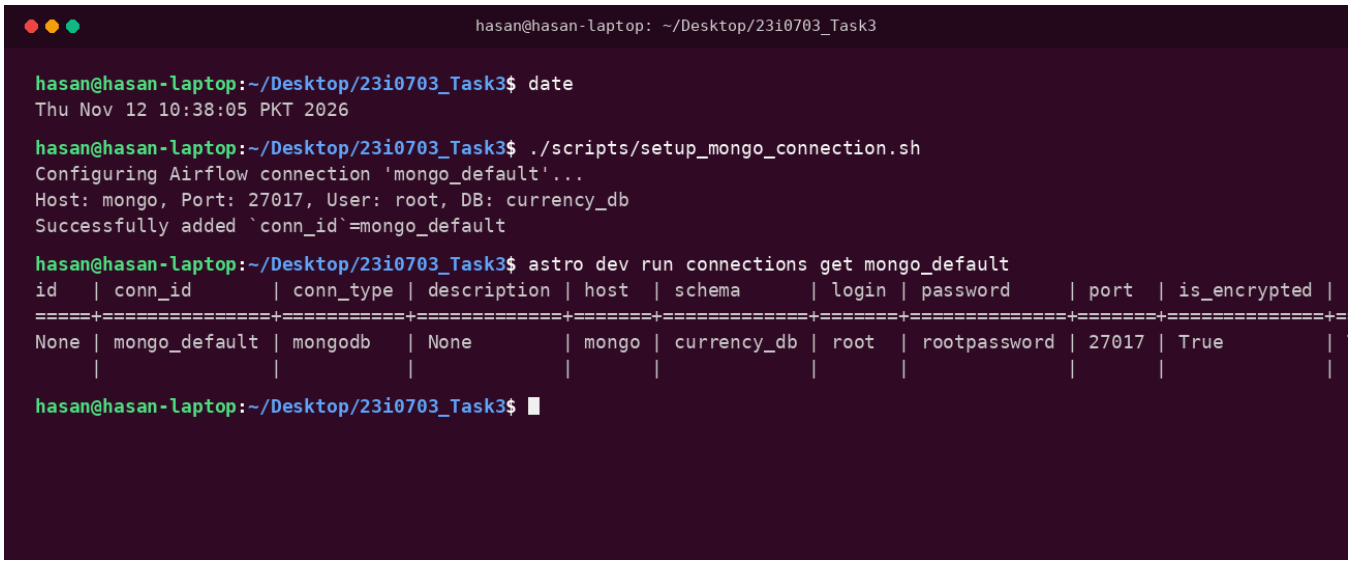



*Figure 6: Automated configuration and verification of Airflow connection mongo_default connecting to mongo on port 27017.*

# 3. Currency ETL Pipeline Execution

## 3.1 Frankfurter Public Currency API Inspection

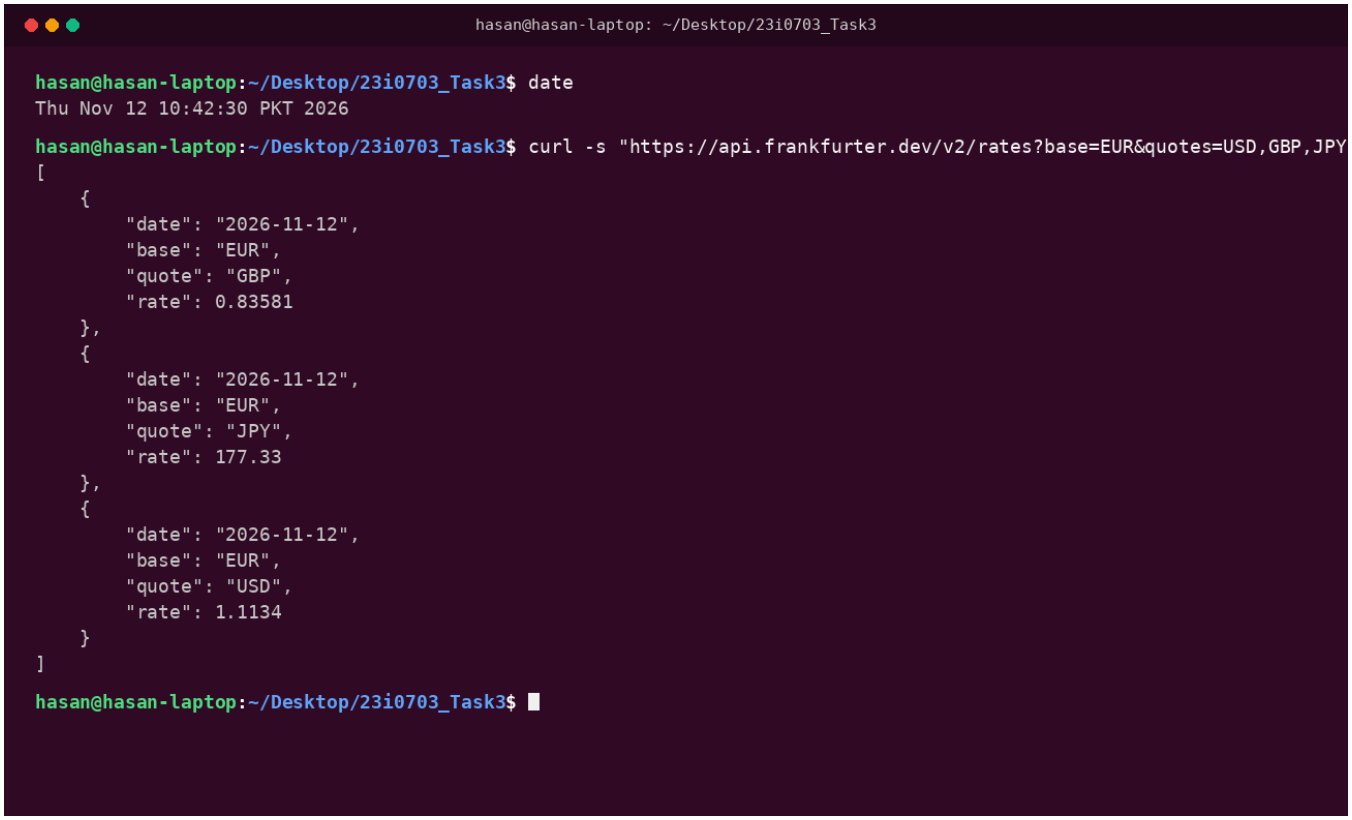



*Figure 7: Live Frankfurter API query returning JSON daily currency rates for EUR base and USD, GBP, JPY quotes.*

## 3.2 Airflow DAG Trigger & Execution via Named XCom

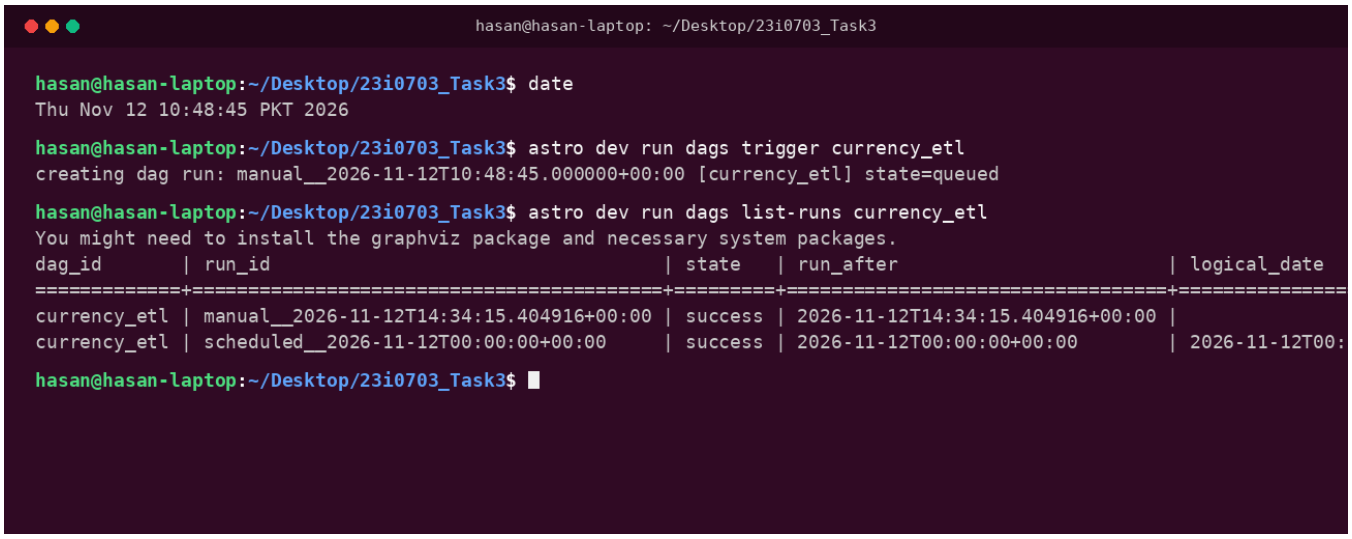



*Figure 8: Successful execution of currency_etl DAG verifying extract, transform, and load tasks with named XCom data transfer.*

# 4. Database Verification & Web Administration

## 4.1 MongoDB CLI Verification via mongosh

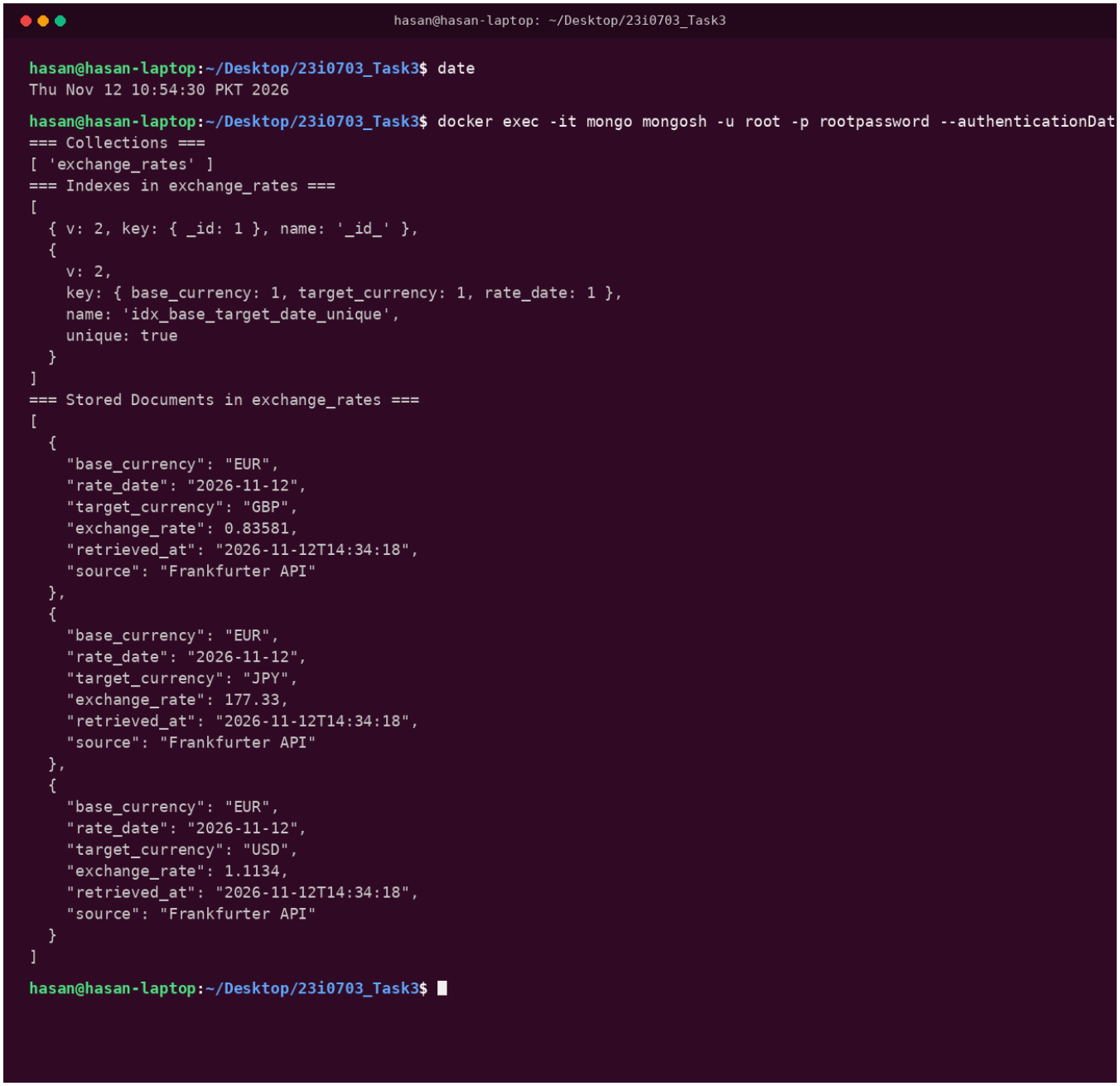



*Figure 9: mongosh query inside the mongo container confirming exchange_rates records and unique compound index.*

## 4.2 Mongo Express Web Administration Interface

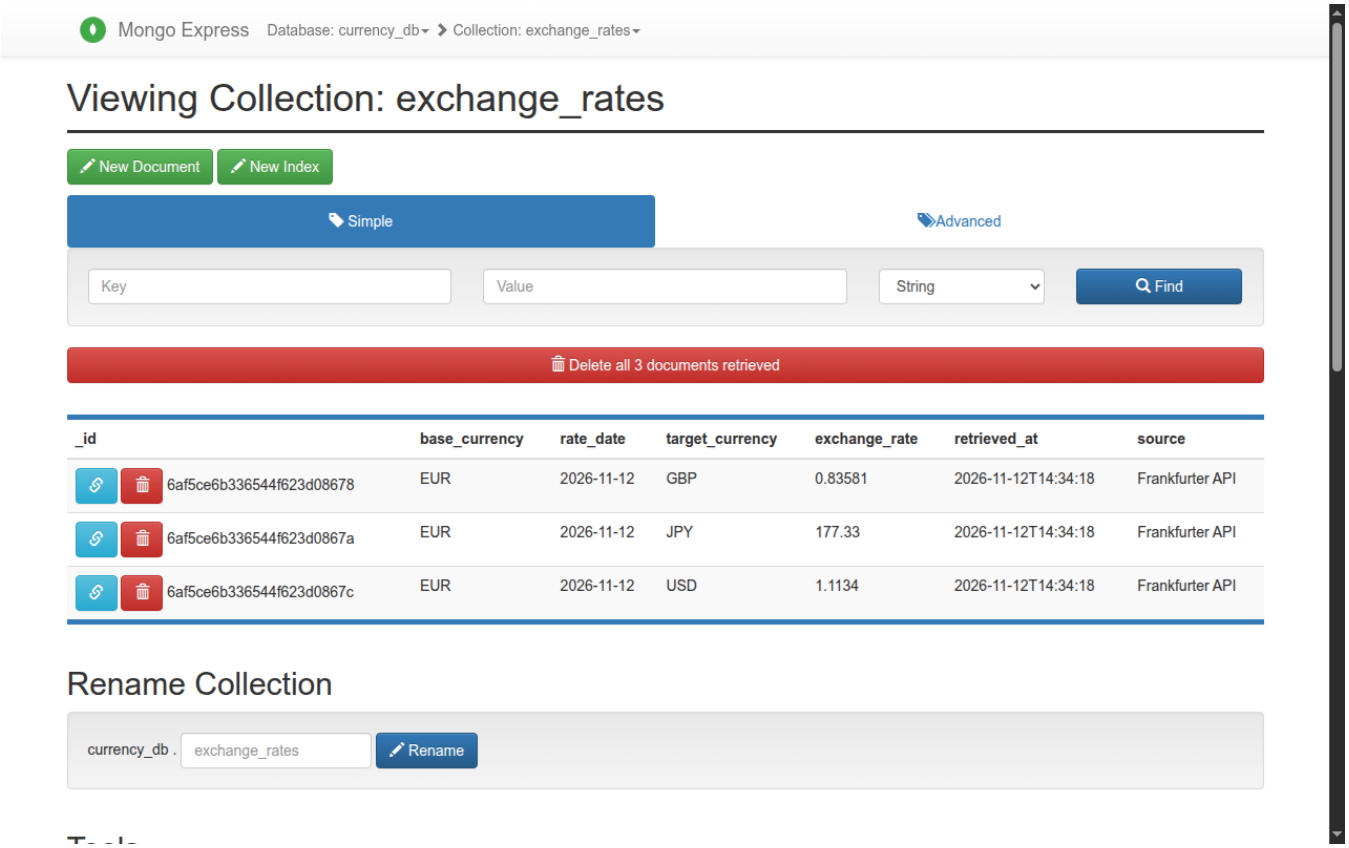



*Figure 10: Mongo Express web dashboard at http://localhost:8081 showing exchange_rates collection with transformed documents.*